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On the left in the address field, press the **ENTER key** to start a new line.

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Division/Dept.: Business Enablement & IT

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.NET development hands-on assessment

We are pleased to present a coding challenge tailored for a .NET Software Engineer. This assessment assumes familiarity with ASP.NET Core and REST APIs.

Submission Guidelines:
You are required to submit your project or a link to your work within 7 days of receiving this assessment to proceed with the next steps in our evaluation process.

Development Standards:
You are encouraged to utilize any design patterns and error handling methodologies that align with industry best practices. Please ensure that your implementation adheres to coding standards typically employed in professional projects. You are welcome to enhance the specified tasks with additional features that you believe would improve the overall quality of the project. Should you have any questions, please do not hesitate to reach out via email.

Project Overview:
Your task is to develop an API that manages products, exposing RESTful endpoints that provide standard CRUD functionality for the Product entity.

CRUD Endpoints

- **GET** /api/products
- **POST** /api/products
- **GET** /api/products/{id}
- **PUT** /api/products/{id}
- **DELETE** /api/products/{id}

Additional Endpoints

- **POST** /api/products/{id}/decrement-stock/{quantity}
  This endpoint should return an OK response after decrementing the stock by the specified quantity.
- **POST** /api/products/{id}/add-to-stock/{quantity}
  This endpoint should return an OK response after incrementing the stock by the specified quantity.
- **GET** /api/products/search?name={name}
  This endpoint should allow searching for products by name, supporting partial matches.
- **GET** /api/products/stock-level?min={min}&max={max}
  This endpoint should return products within a specified stock range.

Requirements:
- The product identifier should be auto-generated as a unique 6-digit number. Even when using multiple instances at the same time, no duplicate IDs should be generated.
- Implement validation for product creation and updates, ensuring that all required fields are present and have valid values.
- Every product must possess a unique identifier. When returning a list of products from any of the aforementioned endpoints, the available stock for each product must be included in the response.

Database Requirements:
Please create a database using Entity Framework Core (EFCore) migrations with a code-first approach. The database schema supports the additional fields and relationships you define.Implement database seeding to populate the database with initial data for testing.

Self-Check List Before Submission

☐ Ensure that your code is clean, well-formatted, executable, and testable.
☐ Apply best practices and design patterns where appropriate.
☐ Confirm that all 9 required endpoints are implemented.
☐ Include logic for product ID generation suitable for a distributed environment with multiple instances.
☐ Utilize EF migrations with a code-first approach to create the database.
☐ Provide unit test cases for various scenarios.
☐ Share your code along with documentation for running the project locally.

Optional: Behavior-Driven Development (BDD) tests to validate expected user outcomes.

We look forward to reviewing your submission!

Best Regards,

Carl Zeiss Vision International
Business Enablement & IT Team